Quantum Computation: Assignment 3

Name: _____ Date: October 09, 2026

Question 1

Convention. Qiskit orders qubits little-endian: a two-qubit basis ket is written |q1 q0⟩, so qubit 0 is the *rightmost* bit and the rightmost factor of every Kronecker product. A gate G on qubit 0 is therefore I ⊗ G, and on qubit 1 it is G ⊗ I. The basis is ordered |00⟩, |01⟩, |10⟩, |11⟩ (rows/columns 0–3). Every circuit starts in ψ_i = |00⟩. Each hand-derived matrix below was checked against `qiskit.quantum_info.Operator(qc)` (within floating-point precision), and each circuit was sampled with 1024 shots on `AerSimulator` (seed 2026). Single-qubit matrices used: I = [[1,0],[0,1]], X = [[0,1],[1,0]], Y = [[0,−i],[i,0]], Z = [[1,0],[0,−1]], H = (1/√2)[[1,1],[1,−1]].

How the calculations are obtained

For A = [[a,b],[c,d]], the Kronecker product replaces every scalar entry with a scaled copy of B: A ⊗ B = [[aB,bB],[cB,dB]]. Thus two 2×2 matrices give one 4×4 matrix. For successive gates, the later gate multiplies on the left. In these four circuits, gates on different qubits combine as (G1 ⊗ I)(I ⊗ G0) = G1 ⊗ G0.

The ordered basis vectors are |00⟩ = $(1,0,0,0)^T$, |01⟩ = $(0,1,0,0)^T$, |10⟩ = $(0,0,1,0)^T$, and |11⟩ = $(0,0,0,1)^T$. Multiplying U by a basis vector selects its corresponding column. Equivalently, (A ⊗ B)(|a⟩ ⊗ |b⟩) = A|a⟩ ⊗ B|b⟩; the action tables below show both methods through the product step and final ket.

Useful identities are X|0⟩ = |1⟩, X|1⟩ = |0⟩; Y|0⟩ = i|1⟩, Y|1⟩ = −i|0⟩; Z|0⟩ = |0⟩, Z|1⟩ = −|1⟩; and H|0⟩ = |+⟩ = (|0⟩ + |1⟩)/√2, H|1⟩ = |−⟩ = (|0⟩ − |1⟩)/√2.

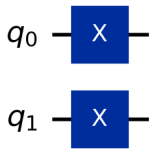
Born rule. Write |ψf⟩ = a00|00⟩ + a01|01⟩ + a10|10⟩ + a11|11⟩. The amplitude for outcome b is ab = ⟨b|ψf⟩, and its probability is P(b) = |ab|² = ab*ab. Normalization requires the four probabilities to sum to 1. The predictions below use the default Qiskit input |00⟩. Actions on the other three basis inputs are included separately, as the assignment requests.

For N independent shots, the expected count is NP(b), and the marginal count standard deviation is √(NP(b)(1−P(b))). With N = 1024, a probability of 1/4 gives an expected 256 counts with a standard deviation of about 13.9. An ideal simulator's observed frequencies need not equal the exact probabilities; zero-probability outcomes remain absent. All four outcome labels are retained in each histogram.

The bit order and controlled-gate convention follow IBM's Qiskit bit-ordering documentation (accessed Oct. 9, 2026).

Question 1(a)

```
qc = QuantumCircuit(2)
qc.x(0)
qc.x(1)
```



4×4 operator (Kronecker product). `qc.x(0)` flips q0 and `qc.x(1)` flips q1. The two gates act on different qubits, so they form one layer and the 4×4 operator is the Kronecker product $U = X_{q1} \otimes X_{q0} = X \otimes X$. Writing $X \otimes X$ in 2×2 blocks, $X \otimes X = [[0 \cdot X, 1 \cdot X], [1 \cdot X, 0 \cdot X]]$, gives an anti-diagonal of ones:

$$U = X \otimes X = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

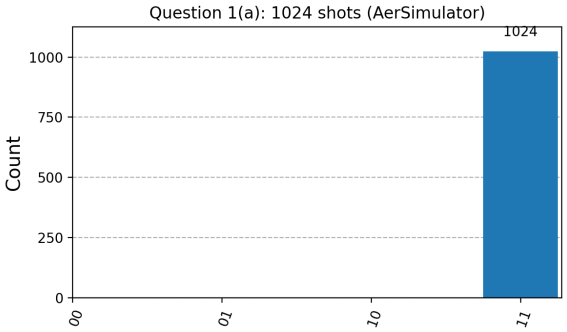


Figure 1(a). Qiskit histogram, 1024 shots (measure_all() appended to the circuit shown above).

Action on all four basis kets. Each row is the corresponding column of U written as a ket. Apply each gate to its own qubit, then expand the product.

Input	Product rule step (q1 ⊗ q0)	U |input⟩
|00⟩	X|0⟩ ⊗ X|0⟩	|11⟩
|01⟩	X|0⟩ ⊗ X|1⟩	|10⟩
|10⟩	X|1⟩ ⊗ X|0⟩	|01⟩
|11⟩	X|1⟩ ⊗ X|1⟩	|00⟩

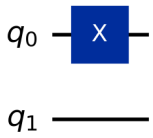
Measurement prediction for |00⟩. The output is $\psi_f = U|00\rangle = |11\rangle$. Both qubits are flipped, so the circuit deterministically outputs |11⟩.

Outcome b	Amplitude ⟨b|ψf⟩	P(b) = |amplitude|²	Expected count	Observed count (fraction)
00	0	0	0	0 (0.000)
01	0	0	0	0 (0.000)
10	0	0	0	0 (0.000)
11	1	1	1024	1024 (1.000)

For any other basis input, square the magnitudes in its action-table row. The one nonzero coefficient has magnitude 1, so that row's output ket occurs with certainty.

Question 1(b)

```
qc = QuantumCircuit(2)
qc.x(0)
```



4×4 operator (Kronecker product). Only q0 receives a gate; q1 is left alone, which is the identity. Hence $U = I_{q1} \otimes X_{q0} = I \otimes X = [[1 \cdot X, 0 \cdot X], [0 \cdot X, 1 \cdot X]]$, i.e. X repeated down the block diagonal:

$$U = I \otimes X = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

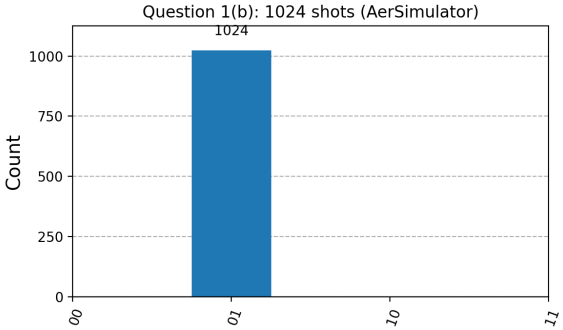


Figure 1(b). Qiskit histogram, 1024 shots (measure_all() appended to the circuit shown above).

Action on all four basis kets. Each row is the corresponding column of U written as a ket. Apply each gate to its own qubit, then expand the product.

Input	Product rule step (q1 ⊗ q0)	U |input⟩
|00⟩	I|0⟩ ⊗ X|0⟩	|01⟩
|01⟩	I|0⟩ ⊗ X|1⟩	|00⟩
|10⟩	I|1⟩ ⊗ X|0⟩	|11⟩
|11⟩	I|1⟩ ⊗ X|1⟩	|10⟩

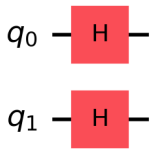
Measurement prediction for |00⟩. The output is $\psi_f = U|00\rangle = |01\rangle$. Only q0 is flipped. In Qiskit's bit-string convention (q1 on the left) the outcome is **01**, not 10.

Outcome b	Amplitude ⟨b|ψf⟩	P(b) = |amplitude|²	Expected count	Observed count (fraction)
00	0	0	0	0 (0.000)
01	1	1	1024	1024 (1.000)
10	0	0	0	0 (0.000)
11	0	0	0	0 (0.000)

For any other basis input, square the magnitudes in its action-table row. The one nonzero coefficient has magnitude 1, so that row's output ket occurs with certainty.

Question 1(c)

```
qc = QuantumCircuit(2)
qc.h(0)
qc.h(1)
```



4×4 operator (Kronecker product). A Hadamard acts on each qubit, so $U = H_{q1} \otimes H_{q0} = H \otimes H = (1/\sqrt{2})[[H, H], [H, -H]]$. Each entry is $(1/\sqrt{2})(1/\sqrt{2})(\pm 1) = \pm\tfrac{1}{2}$, and the sign is $(-1)^{x \cdot y}$ (bitwise dot product of row and column labels):

$$U = H \otimes H = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

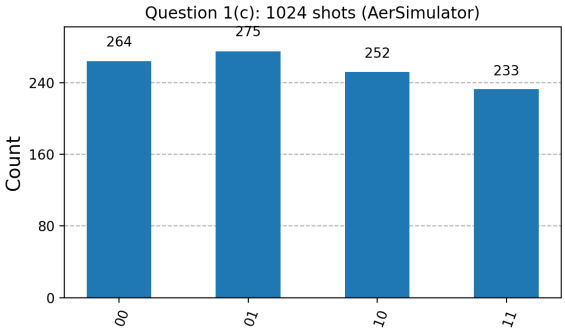


Figure 1(c). Qiskit histogram, 1024 shots (measure_all() appended to the circuit shown above).

Action on all four basis kets. Each row is the corresponding column of U written as a ket. Apply each gate to its own qubit, then expand the product.

Input	Product rule step (q1 ⊗ q0)	U |input⟩
|00⟩	H|0⟩ ⊗ H|0⟩ = |+⟩ ⊗ |+⟩	½(|00⟩ + |01⟩ + |10⟩ + |11⟩)
|01⟩	H|0⟩ ⊗ H|1⟩ = |+⟩ ⊗ |−⟩	½(|00⟩ − |01⟩ + |10⟩ − |11⟩)
|10⟩	H|1⟩ ⊗ H|0⟩ = |−⟩ ⊗ |+⟩	½(|00⟩ + |01⟩ − |10⟩ − |11⟩)
|11⟩	H|1⟩ ⊗ H|1⟩ = |−⟩ ⊗ |−⟩	½(|00⟩ − |01⟩ − |10⟩ + |11⟩)

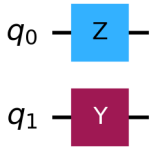
Measurement prediction for |00⟩. The output is $\psi_f = U|00\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. H ⊗ H maps |00⟩ to an equal superposition of all four basis states, so each outcome has probability ¼. With 1024 shots each bar should be close to 256; the spread seen in the histogram is ordinary shot noise (standard deviation $\sqrt{1024 \cdot \tfrac{1}{4} \cdot \tfrac{3}{4}} \approx 13.9$ counts).

Outcome b	Amplitude ⟨b|ψf⟩	P(b) = |amplitude|²	Expected count	Observed count (fraction)
00	½	1/4	256	264 (0.258)
01	½	1/4	256	275 (0.269)
10	½	1/4	256	252 (0.246)
11	½	1/4	256	233 (0.228)

For any other basis input, square the magnitudes in its action-table row. Every input in part (c) gives probability 1/4 for every outcome; the signs affect phases.

Question 1(d)

```
qc = QuantumCircuit(2)
qc.z(0)
qc.y(1)
```



4×4 operator (Kronecker product). Z acts on q0 and Y on q1. Because they act on different qubits the order in which they are written does not matter, and $U = Y_{q1} \otimes Z_{q0} = Y \otimes Z = [[0 \cdot Z, -i \cdot Z], [i \cdot Z, 0 \cdot Z]]$:

$$U = Y \otimes Z = \begin{bmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{bmatrix}$$

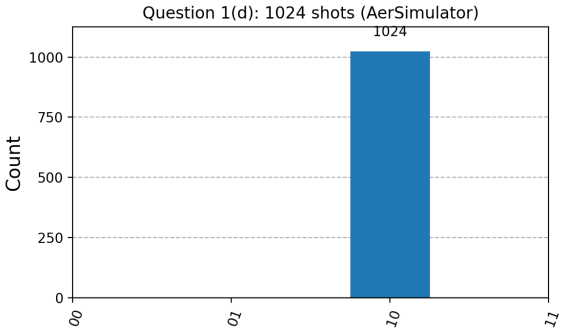


Figure 1(d). Qiskit histogram, 1024 shots (measure_all() appended to the circuit shown above).

Action on all four basis kets. Each row is the corresponding column of U written as a ket. The middle column explicitly multiplies the Y and Z phases.

Input	Product rule step (q1 ⊗ q0)	U |input⟩
|00⟩	(i|1⟩) ⊗ |0⟩	i|10⟩
|01⟩	(i|1⟩) ⊗ (−|1⟩)	−i|11⟩
|10⟩	(−i|0⟩) ⊗ |0⟩	−i|00⟩
|11⟩	(−i|0⟩) ⊗ (−|1⟩)	i|01⟩

Measurement prediction for |00⟩. The output is $\psi_f = U|00\rangle = i|10\rangle$. Z leaves |0⟩ unchanged and $Y|0\rangle = i|1\rangle$, so $\psi_f = i|10\rangle$. The global phase i is not observable: $|\langle 10|\psi_f\rangle|^2 = |i|^2 = 1$.

Outcome b	Amplitude ⟨b|ψf⟩	P(b) = |amplitude|²	Expected count	Observed count (fraction)
00	0	0	0	0 (0.000)
01	0	0	0	0 (0.000)
10	i	1	1024	1024 (1.000)
11	0	0	0	0 (0.000)

For any other basis input, square the magnitudes in its action-table row. The one nonzero coefficient has magnitude 1, so that row's output ket occurs with certainty.

Question 2

Read Figure 1 from left to right. After the initial barrier, H acts on q0. The filled dot on q0 is connected vertically to the X on q1: together they denote a **controlled-X (CNOT)**, implemented by `qc.cx(0, 1)`. q0 is the control and q1 the target. A second barrier precedes measurement of q0 into c0 and q1 into `c1`. Barriers constrain compilation but leave the state unchanged.

In the basis |q1 q0⟩, CNOT maps |q1 q0⟩ to |q1 XOR q0, q0⟩: |00⟩ → |00⟩, |01⟩ → |11⟩, |10⟩ → |10⟩, and |11⟩ → |01⟩. It acts coherently on a superposition. Starting from |ψ⟩ = |00⟩:

Step	Operator	State
Input	—	|00⟩
H on q0	I ⊗ H	(1/√2)(|00⟩ + |01⟩)
CX(0, 1)	C	(1/√2)(|00⟩ + |11⟩)

The controlled operator can be written using projectors on the control: C = I ⊗ |0⟩⟨0| + X ⊗ |1⟩⟨1|. The target is unchanged when q0 = 0 and flipped when q0 = 1. The total operator is U = C(I ⊗ H), with the first gate on the right. Expanding these products gives:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad U = 1/\sqrt{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

The first column of U is U|00⟩. Equivalently, C leaves the |00⟩ branch unchanged and maps the |01⟩ branch to |11⟩. Thus the output state vector **just before measurement** is

$$\mathbf{|\psi_{out}\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle) = (1/\sqrt{2}, 0, 0, 1/\sqrt{2})^T = |\Phi+\rangle}$$

This is an entangled Bell state: each qubit individually has equal 0/1 probabilities, while their results always agree. It cannot factor as (a|0⟩ + b|1⟩) ⊗ (c|0⟩ + d|1⟩): nonzero 00 and 11 coefficients would require all four factors to be nonzero, which would also make the 01 and 10 coefficients nonzero. The reduced state of either qubit is I/2.

Question 2: measurement verification

Applying the Born rule to the output vector gives P(00) = $|1/\sqrt{2}|^2$ = 1/2, P(11) = 1/2, and P(01) = P(10) = 0. The counts use the classical bit order c1c0. With 1024 shots, each allowed outcome has an expected 512 counts and a marginal standard deviation $\sqrt{(1024 \cdot ½ \cdot ½)}$ = 16 counts.

Outcome c1c0	Predicted	Simulated (1024 shots)
00	1/2	516 (0.504)
01	0	0 (0.000)
10	0	0 (0.000)
11	1/2	508 (0.496)

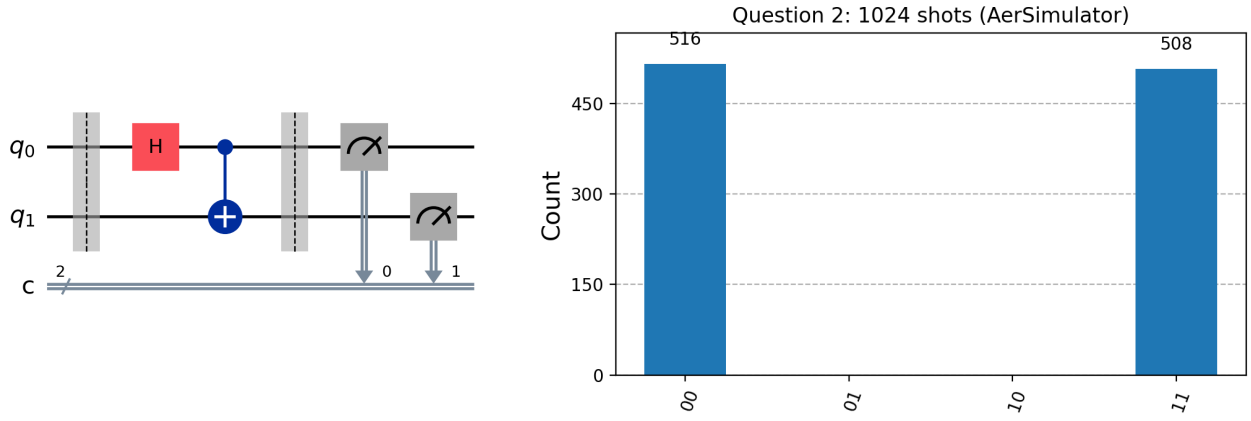


Figure 2. Question 2 circuit rebuilt in Qiskit and its simulated histogram (about 50/50 between 00 and 11).

State after measurement. In a single ideal run the state collapses to |00⟩ or |11⟩, each with probability 1/2. If the outcome record is discarded, the ensemble is ρ = ½|00⟩⟨00| + ½|11⟩⟨11| = diag(½,0,0,½), rather than one pure state vector. The Bell vector above describes the coherent state before the measurement gates.

The histogram verifies the predicted computational-basis probabilities. Establishing entanglement experimentally would also require measurements in other bases: a classical 50/50 mixture of |00⟩ and |11⟩ gives this same histogram. Here the statevector calculation verifies the ideal circuit's coherent Bell state.

Question 3: How qubits are represented on an IBM quantum computer

IBM represents each physical qubit using a superconducting *transmon*: an electrical circuit with a Josephson junction connected in parallel with a capacitor [1]. The junction contains a thin insulating barrier between superconductors and supplies a nonlinear inductance. Together with the capacitance, it produces a quantum oscillator with unequal energy-level spacings. Its ground and first excited energy eigenstates define the computational states $|0\rangle$ and $|1\rangle$ [2, 3]. Within this two-level subspace, a pure state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with complex amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$; their relative phase also carries quantum information. The large capacitance reduces sensitivity to charge noise, while the remaining anharmonicity permits selective control [2]. IBM cools its processors in dilution refrigerators to temperatures of roughly 15–20 mK to reduce thermal excitation [4].

Shaped microwave pulses manipulate the amplitudes and relative phase to implement single-qubit gates, while interactions between coupled qubits implement entangling gates [1, 3]. A transmon has higher energy levels, so pulse calibration and shaping suppress unwanted population leakage outside $|0\rangle$ and $|1\rangle$; leakage is not automatically zero [3]. For dispersive readout, a transmon couples to a microwave resonator whose response depends on the qubit state. A probe signal is amplified and classified to return a classical 0 or 1 [1, 3]. Ideally, a computational-basis measurement of $\alpha|0\rangle + \beta|1\rangle$ returns 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$. One such measurement does not reveal both complex amplitudes; repeated preparations and measurements estimate probabilities. Physical devices also experience relaxation, dephasing, control imperfections, and readout errors [3]. The histograms in this assignment instead use an ideal AerSimulator without a hardware noise model: their random variation comes from finite-shot sampling of the predicted probabilities.

References

[1] IBM Quantum Learning, “Running quantum circuits,” IBM Quantum course (accessed Oct. 9, 2026).

[2] J. Koch *et al.*, “Charge-insensitive qubit design derived from the Cooper pair box,” *Phys. Rev. A* 76, 042319 (2007). doi:10.1103/PhysRevA.76.042319.

[3] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, “Circuit quantum electrodynamics,” *Rev. Mod. Phys.* 93, 025005 (2021). doi:10.1103/RevModPhys.93.025005.

[4] IBM Quantum, “Five years ago today, we put the first quantum computer on the cloud. Here’s how we did it,” May 4, 2021. IBM Quantum blog (accessed Oct. 9, 2026).

Appendix: Qiskit code used for verification

Full source: `quantum_hw3.py` (circuits, operator check, simulation, figures) and `make_report.py` (this PDF). Core excerpt:

```
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Operator, Statevector
from qiskit.visualization import plot_histogram
from qiskit_aer import AerSimulator
import numpy as np

X = np.array([[0, 1], [1, 0]])
qc = QuantumCircuit(2)          # Question 1(a)
qc.x(0)
qc.x(1)

# 1) verify the hand-derived Kronecker product  U = X (x) X
assert np.allclose(Operator(qc).data, np.kron(X, X))

# 2) sample the output and plot the histogram
qc.measure_all()
sim = AerSimulator(seed_simulator=2026)
counts = sim.run(transpile(qc, sim), shots=1024).result().get_counts()
plot_histogram(counts)          # -> {'11': 1024}

# Question 2
qc2 = QuantumCircuit(2, 2)
qc2.barrier()
qc2.h(0)
qc2.cx(0, 1)                  # dot on q0 controls the X on q1
qc2.barrier()
psi = Statevector.from_label("00").evolve(qc2)
assert np.allclose(psi.data, np.array([1, 0, 0, 1]) / np.sqrt(2))
qc2.measure([0, 1], [0, 1])
counts2 = sim.run(transpile(qc2, sim), shots=1024).result().get_counts()
plot_histogram(counts2)        # approximately {'00': 512, '11': 512}
```